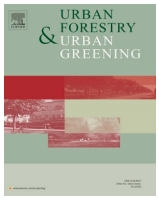
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Original article



Balancing biodiversity conservation and recreational use in nature-oriented parks in Saitama Prefecture, Japan

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ABSTRACT

Urban parks play critical roles in balancing biodiversity conservation and public recreation, yet the multifunctional demands they face frequently create tensions between ecological objectives and recreational uses. In practice, prioritising one goal can constrain the other, and managers must mitigate the resulting disadvantages while conserving urban biodiversity. This study investigates the strategies that park managers use to manage biodiversity conservation–recreation trade-offs across five nature-oriented parks in Saitama Prefecture, Japan (Kamogawa Mizubenosato Park, Sayama Hill Flora & Fauna Interaction Centre, Saitama Green Forest Museum, Kitamoto Nature Observation Park, and Ageo Maruyama Park). These strategies are grouped into three analytically derived orientations: Environmental Focus, Balanced Focus, and User Focus. A mixed-methods approach integrating text mining, co-occurrence network analysis, and thematic content analysis of semi-structured interviews was employed. A novel 30-level Conservation and Utilisation Management Framework was developed that classifies management orientations, defined here as managers' dominant strategic direction for handling conservation–recreation trade-offs. Orientations are associated with external environmental conditions (e.g., land-use context), internal site characteristics (e.g., scale and naturalness), and management perspectives (e.g., policy mandates and partnerships). Across all sites, user communication methods, including regulation, collaboration, and experiential programming—played a central role in reconciling conservation with recreational use. This analytical framework provides a practical and adaptable tool for diagnosing park governance orientations, guiding policy development, and integrating biodiversity conservation and recreational objectives in urban and peri-urban park management.

1. Introduction

1.1. Context

Rapid global population growth has accelerated urban expansion (Seto et al., 2012), significantly reducing green spaces as cities strive to meet demands for housing, infrastructure, and resources (Haaland and Konijnendijk van den Bosch, 2015; Colding et al., 2020). This trend highlights the urgent need for conservation strategies that integrate ecological sustainability and urban development. Two core frameworks have emerged in this context: green infrastructure, referring to strategically planned networks of natural and semi-natural areas that provide environmental, economic, and social benefits (Malik and Ali, 2023), and ecosystem services, which emphasise how ecological processes support human well-being through functions such as air purification, climate regulation, and recreation (Torabi et al., 2020). In this study, urban parks are conceptualized as governance settings in which management strategies shape both ecological functions and public benefits, and in which public understanding of the value of biodiversity can be fostered (Nesbitt et al., 2017; Miller, 2005).

Urban parks, as central elements of green infrastructure, deliver multifunctional value. They provide psychological, social, and health benefits (Kaoma and Shackleton, 2014; Haq, 2011), while also supporting biodiversity, stress mitigation, and ecological corridors (Nesbitt et al., 2017; Ugolini et al., 2018; Liu et al., 2018). However, they also face the persistent challenge of balancing recreational use with environmental conservation. Here, "balancing" refers to negotiating conservation–recreation trade-offs (and, where feasible, synergies) through context-sensitive management instruments that protect ecological values while maintaining public benefits (McShane et al., 2011; Young et al., 2016). In this study, management orientation refers to a manager's primary strategic direction for navigating trade-offs along a

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conservation–use spectrum. Specifically, it denotes positioning a park within a 30-level ordered continuum, where each level reflects the relative priority given to environmental protection versus freedom of use.

Research on conservation–recreation coexistence and conflict resolution shows that recreation can degrade sensitive habitats (Ballantyne et al., 2014) and that managing these tensions often depends on governance processes, such as rules, spatial differentiation (Nepal and Weber, 1995; Gobster, 2007), and trust-building (Young et al., 2016). Conflicts emerge from the introduction of non-native species, user safety concerns, and tensions between conservation advocates and recreational users. Overharvesting of cicadas in wetlands, discarded fishing gear in aquatic systems, and restricted-access restoration zones illustrate the difficulties of reconciling ecological protection with diverse user needs.

In aligned with international targets, such as the Kunming-Montreal Global Biodiversity Framework (2022), calling for the conservation 30% of global land and marine areas by 2030, Japan has enacted domestic policies to strengthen local biodiversity. The Act on the Promotion of Activities for Enhancing Biodiversity (Government of Japan, 2024) prioritizes urban green spaces, forests, and coastal zones (Watanabe et al. 2024). This legislation encourages collaboration among public authorities, private stakeholders, and civil society, reaffirming the multifunctional role of parks for localised conservation action.

1.2. Literature review

Despite the increasing recognition of the ecological value of urban parks, conflicts between human activity and wildlife remain complex (Nepal and Weber, 1995; Nyhus, 2016). Recreational activities such as dog walking, sports, and picnicking can degrade habitats, cause biodiversity loss, and contribute to littering (Ballantyne et al., 2014). These impacts are exacerbated by limited ecological awareness among users, insufficient staff training (Schipperijn et al., 2010; Talal and Santelmann, 2020), and spatial constraints that fragment habitats. Addressing these conflicts requires integrated, multi-stakeholder strategies.

Previous studies identify four main approaches.

1. Stakeholder Collaboration

Engaging authorities, researchers, residents, and park users helps reconcile ecological and recreational demands (Haq, 2011; Huang, 2009), supported by trust-building (Young et al., 2016), partnerships with NGOs and government (Scholz and Steiner, 2015), and participatory learning (Ugolini et al., 2018).

2. Governmental Measures

Policies, legislation, and enforcement encourage responsible use and help ensure that development aligns with biodiversity goals (Haq, 2011; Huang, 2009; Nyhus, 2016).

3. Technical Measures

Tools such as zoning, buffer zones, seasonal restrictions, ecological restoration, and biodiversity monitoring enhance resilience (Cascante et al., 2014; Jim, 2004; Akter and Dhar, 2017).

4. Cognitive Approaches

Education, awareness campaigns, and volunteer participation promote long-term stewardship (Gobster, 2007; Miller, 2005).

Despite these advances, no universally adaptable model has been developed. While many frameworks emphasise tailoring strategies to ecological, social, and cultural contexts (Cascante et al., 2014; Eagles et al., 2013; Torabi et al., 2020), they tend to reflect traditional urban park planning and management—amenity-oriented provision, fixed zoning or regulatory approaches grounded in relatively stable missions, resources, and objectives—that leave them ill-equipped to handle changing circumstances and complex stakeholder dynamics. Amid evolving expectations for biodiversity and green infrastructure (e.g., more ambitious biodiversity targets, pressures from invasive species, and shifting GI functions), few guidelines exist for revising the priorities of park functions.

Another concern is the tendency to treat stakeholder involvement as a formal consultation requirement rather than an ongoing negotiation of rights, responsibilities, and legitimacy. In practice, financial constraints have reduced management capacity and shiftied objectives, even as demands from increasingly diverse user groups grow. These pressures manifest in conflicts such as disputes over the nuisance factor of ball games, changing power balances associated with population aging that may disadvantage either older adults or children, barrier-free interventions that inadvertently degrade natural areas, excessive accessibility improvements that threaten habitats, and the neglect of green spaces resulting in the dominance of a single invasive herbaceous species. Consequently, tensions between biodiversity conservation and park use in park management are increasing.

Urban parks are embedded within wider institutional, ecological, and social systems. Recent scholarship emphasises adaptive, place-based approaches reflecting site-specific features, community relations, and management roles (Takyi and Seidel, 2017). Much research quantifies stakeholder perspectives using Likert scales for analytical purposes (Eagles et al., 2013), although generalised questions may overlook context-specific issues critical at the local level. In contrast, studies that integrate the qualitative perspectives of managers and conservation volunteers are limited (e.g., interview/participatory studies: Eagles et al., 2013; Ugolini et al., 2018; and survey-based use studies: Schipperijn et al., 2010), resulting in a gap in understanding the evolutionary processes of governance models.

This study addresses that gap by analysing localised strategies in five urban parks in Saitama Prefecture, Japan, Managers' and stakeholders' narratives are examined to help determine how conservation and recreational needs are balanced.

Methodologically, prior KH Coder applications combine frequency analysis with Jaccard-based co-occurrence mapping and interpret (or refine) co-occurrence patterns by checking keyword usage in the original text (Koga, 2019). This study applies this approach to park-manager interviews, integrating frequency/Jaccard outputs with close reading and thematic coding to support cross-park interpretation.

1.3. Objectives

This study examines five nature-oriented parks in Saitama: three in urban areas, Kamogawa Mizubenosato Park (Kamogawa), Kitamoto Nature Observation Park (Kitamoto), and Ageo Maruyama Park (Maruyama), and a regional park with two independently managed zones: Sayama Hill Flora and Fauna Interaction Centre (Sayama) and Saitama Green Forest Museum (Museum).

Through interviews with park managers and conservation volunteers, text mining, and contextual analysis, the study seeks to determine the factors that shape park managers' management orientations and strategic choices in balancing biodiversity conservation and recreation. To address the lack of comparable cross-park evidence, this study combines interviews, text mining, and the Conservation and Utilisation Management Framework to classify orientations across parks. The study's contribution is primarily empirical: it presents an interview-grounded organising scheme that arranges interview statements along a conservation–utilisation axis for cross-park comparison. The study pursues three objectives:

1. Develop a multi-level framework for classifying park management policies along a spectrum from Environmental Focus to User Focus, with intermediate Balanced orientations.

Measures related to environmental conservation and user convenience in natural environments were identified through interviews conducted in five parks and organised into a multi-level framework structured from an environmental conservation perspective.

2. Apply the framework to analyse how external factors (e.g., rivers, surrounding forest cover, and topography), internal conditions (e.g., naturalness, habitat diversity, and infrastructure), and management factors (e.g., policy mandates, institutional arrangements, and manager perspectives) shape management orientations, using the Conservation

and Utilisation Management Framework (Fig. 1).

This objective tests the hypothesis that the relative levels of management measures along the conservation–use spectrum are shaped by internal and external environmental conditions as well as management factors.

3. Identify strategies for reconciling biodiversity conservation with recreational use across urban and regional park contexts (management and governance strategies).

2. Methods

2.1. Study area

Interviews were conducted with those responsible for managing and conserving five parks, referred to as "managers." At Sayama, Museum, and Kitamoto, designated managers were interviewed, whereas at Kamogawa and Maruyama, respondents were active volunteers engaged in habitat restoration.

Sayama and Museum are regional parks in the Sayama Hills, a corridor that connects fragmented natural habitats between the urban and suburban areas of Saitama and Tokyo. This landscape, a mosaic of secondary forests, grasslands, and wetlands, supports high biodiversity and facilitates wildlife movement and genetic exchange, maintaining essential ecological connectivity amid increasing urbanisation. The two sites differ in context: Sayama borders residential neighborhoods, while the Museum is more isolated and surrounded by forested land, illustrating how external settings influence management approaches.

Three urban parks were included to capture contrasts in size, governance, and ecological–recreational balance. Kamogawa park is located within a riparian area associated with a stormwater drainage channel converted from a former river course. Located along a branch of the Arakawa River, it retains a relatively natural setting yet is adjacent to dense residential areas, fostering strong community engagement. Kitamoto, in a suburban urbanisation control zone surrounded by farmland and natural areas, is managed by the Saitama Ecosystem Conservation Society, a foundation specialising in biodiversity protection and environmental education, with a focus on wetland and species conservation; it also operates as a regional centre for environmental education. Maruyama has experienced conflicts between user activities (e.g., fishing) and conservation efforts, especially during pond restoration.

These five parks represent diverse combinations of ecological conditions, spatial infrastructure, user dynamics, and management frameworks. This diversity allows for the examination of how internal, external, and management factors, ranging from park size, ownership, and ecological sensitivity to volunteer involvement and professional management, shape conservation–utilisation strategies in both urban and regional park contexts.

Table 1 summarises the parks' essential attributes, including park name, area, park type, related organisations involved in nature restoration, the interviewee's role, and surrounding land conditions.

2.2. Framework development

A 30-level Conservation and Utilisation Management Framework was derived by categorising interview statements according to their emphasis on environmental protection, public use, or a balance of both. Interview statements were used to elaborate a pre-conceived Environmental Focus–Balanced–User Focus framework, structured as a 30-level coding ladder. Although atypical, the 30-level representation is intentionally fine-grained to retain case-derived distinctions and to offer adaptable reference points for other parks. The levels indicate stepwise shifts in conservation stringency along the conservation–use continuum, rather than a normative "better–worse" ranking of adjacent items. Conservation strictness is the primary ordering principle of the framework, with more regulatory measures indicating a stronger environmental orientation. The presence or absence of activity restrictions is the key criterion for distinguishing between conservation priority and freedom of use. The framework was broadly organised into three bands, each comprising 10 levels: User Focus, Balanced, and Environmental Focus:

- User Focus (Levels 1–10): Prioritising visitor convenience, recreation, and amenity features.
- Balanced (Levels 11–20): Harmonising public use with conservation concerns.
- Environmental Focus (Levels 21–30): Emphasising biodiversity, ecological integrity, and habitat preservation. Each level is linked to characteristic keywords and themes derived from interview data.

The wording of each level was intentionally derived from empirically grounded insights obtained through practitioner interviews, preserving context-specific meanings rather than imposing generalised terminology.

Interview transcripts were segmented into paragraphs, each treated as a "statement" and assigned to one of the 30 content-based levels. Statements falling outside the three categories were excluded. For cross-park comparisons, a proportional index was calculated for each park to summarise the relative emphasis on Environmental Focus, Balanced, and User Focus categories:

$$\text{Proportional value} = (\text{Category Score}/\text{Total Statements}) \times 100\% \quad (1)$$

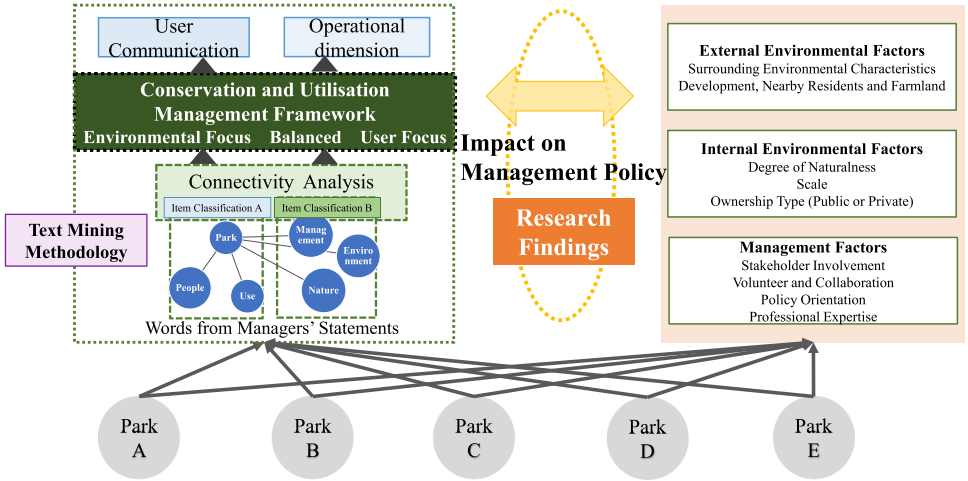


Fig. 1. Overall structure of the study.

Table 1
Summary of park features, management, and context.

Park Name (Abbreviated name here)	Area (ha)	Park Type	Related Organisations for Nature Restoration (interviewee affiliation)	Role of the Interviewee	Organisational expertise (keywords)	Surrounding Land Conditions
Kamogawa Mizubenosato Park (Kamogawa)	1.59	Urban park (municipal)	Kamogawa Mizube Wakuwaku Network (community-based organisation collaborating with the city government)	an advisor from NPO eco-works collaborating with community-based organisation, designated manager and city government	grassland management; habitat creation; nature observation events	Adjacent to residential areas and the Kamogawa River Rural-urban fringe
Sayama Hill Flora and Fauna Interaction Centre (Sayama)	48.5 (Centre area)	Regional Park (prefectural, total designated area:1807.8 ha)	Totoro no Furusato Foundation (land trust organisation)	a representative of the designated management organisation	national-trust conservation; satoyama woodland management; biodiversity surveys	Adjacent to residential areas a satoyama landscape dominated by secondary forest, Rural-urban fringe
Saitama Green Forest Museum (Museum)	85.5 (Museum area)	Located within the same regional park as Sayama	Centre for Environmental Studies (private environmental consulting firm)	a representative of the designated management organisation	nature interpretation; environmental education; satoyama management	Located within the Sayama Hills, featuring a high natural environment with a satoyama landscape of secondary forest
Kitamoto Nature Observation Park (Kitamoto)	32.9	Urban Park (prefectural, Exceptionally focused on nature conservation)	Ecosystem Conservation Society - Saitama (professional conservation organisation)	a representative of the designated management organisation	biodiversity conservation; environmental education; volunteer-supported management	Adjacent to the Arakawa River, farmland, and a hospital, a traditional satoyama valley landscape and wetland environments
Ageo Maruyama Park (Maruyama)	14.9	Urban park (municipal)	NPO eco-works (Seitaikobo, collaborating with city government) Ageo Mizubemori (waterfront volunteers)	a specialist ecological restoration organisation contracted to undertake restoration; coordinator Kaibori (pond draining) and restoration with the city government	kaibori restoration; invasive species control; biodiversity monitoring; collaboration support	Adjacent to the Arakawa River and residential areas featured by rural landscape (wetland and farmland)

where Category Score is the number of statements in a category, and Total Statements is the total number of statements across all levels. Results are presented in Fig. 2 as a stacked bar chart, illustrating each park's relative emphasis on conservation versus public utilisation. This framework enables systematic, comparative analysis while preserving the differences in management orientations across parks.

2.3. Orientation classification

To interpret the parks' management orientations, operational dimensions (Table 2), which specify core management practices, and user communication strategies (Table 3), which guide visitor behaviour, were defined. The items classified in these tables were analyzed according to whether they supported an Environmental Focus, Balanced, or User Focus orientation, reflecting the strategic positions adopted in park management in response to internal and external conditions.

To identify strategies that reconcile biodiversity conservation with recreational use, interview content was classified across five operational dimensions as summarized in Table 2.

Methodologically, each operational dimension was organized by its regulatory and spatial scope: measures affecting broader spatial areas, such as zoning, were assigned to higher regulatory levels (reflecting a stronger environmental orientation), while supporting measures were assigned to lower levels.

Table 2
Operational dimension and definitions.

Operational dimension	Definition
Zoning and Land Use	Decisions on spatial organisation and designated areas for conservation or public use
Management Practices	Maintenance schedules, ecological restoration efforts
Rules and Signage	Regulations and informational guidance for park users
Hardware and Facility Development	Physical infrastructure and facility improvements
Experiential Activities	Programmes and activities engaging visitors

Table 3
User communication classification and definitions.

User Communication	Definition
Regulations	Rules or guidelines to manage park use
Collaboration	Engaging users in joint activities with park managers
Information Dissemination	Providing guidance to ensure appropriate visitor behaviour
Laissez-faire	Minimal intervention; user interactions develop organically

Complementing the implementation of measures classified under the operational dimensions, user communication represents the types of management approaches used to guide or influence visitor behaviour. Based on the interview transcripts, communication practices were classified according to how managers convey messages to users or engage in dialogue when addressing conservation–use tensions in park management. These approaches include one-way communication from administrators, interactive dialogue with users, and decisions that intentionally allow user activities without direct regulation. Accordingly, user communication practices were categorised into four strategies: regulations, collaboration, information dissemination, and laissez-faire (Table 3).

These component classifications were integrated to construct the comprehensive framework (Table 5), serving as the baseline for evaluating how managers balance visitor engagement with ecological conservation along the conservation–use continuum (Fig. 1).

Through this dual-classification approach, the operational dimensions in Table 2 identify the types of management measures, while the user communication strategies in Table 3 indicate how these measures are practically implemented through interactions with users.

Therefore, user communication provides a more practice-oriented basis for understanding how managers address conservation–use conflicts in park management.

2.4. Interviews

Under Japan's designated management system, local governments oversee policy targets, while appointed managers of the parks handle operations, ensuring the translation of policy into practice. This is a qualitative interview study, not a questionnaire survey or statistical analysis. Since environmental outsourcing precludes specialised answers from the administration, expert respondents from entrusted/commissioned restoration organisations were selected for Kamogawa and Maruyama, rather than designated managers; their accounts reflect implementation- and collaboration-oriented perspectives within the governance system.

The interviews were conducted between December 2019 to November 2020, with each session lasting approximately 90 min. The semi-structured interviews covered 24 thematic items (Table A.2) and were conducted in an interactive format to guide discussion toward management themes related to environmental conservation and park use. For example, under the theme of "responses to hazardous species," participants were asked: "If evidence of potentially hazardous species, such as a wasp nest, is found within the park, would it be removed to ensure visitor safety, or retained for environmental conservation purposes? In either case, would additional measures be taken for the benefit of either the species or park users?"

The interviewer engaged respondents in open dialogue and probed further whenever discussions touched on whether management decisions leaned toward stronger environmental protection or greater user orientation. During the interviews, the interviewer probed further to obtain more detailed information and a deeper understanding of the respondents' reasoning. These concepts formed the conceptual basis of the subsequent orientation classification.

Prior to this study, one of the authors conducted coursework-based surveys on environmental education and habitat management with university students at two of the study sites,Kitamoto for six years and Sayama for four years. Through these educational and field-based activities, a range of practical challenges associated with adaptive habitat management, facility development, and environmental education in park green spaces were identified.

The interview questions were developed from coursework discussions that addressed these challenges by organising facility elements and human activities related to recreational park use and examining their potential influences on adaptive habitat management.

Within the 30-level framework, narrative descriptions related to these themes were evaluated to determine whether they could be classified into the pre-established orientations of Environmental Focus, Balanced, or User Focus. Furthermore, the framework's operational dimensions were derived by grouping all descriptions extracted from the orientation-based analysis.

All questions were posed consistently across interviews to ensure analytical rigor. To facilitate subsequent text mining, core keywords were systematically extracted from the responses. Conversations were audio-recorded and transcribed for analysis.

Two academic researchers, both authors of this paper and present during the interviews, specialising in Urban Landscape Planning and Environmental Management, analysed the transcripts in two stages. First, the first author reviewed the textual data line by line, classifying each segment according to whether the primary tone concerned environmental conservation, park use, the relationship between the two, or other topics, and extracted relevant keywords. The second author then reviewed the interpretive results. Any differences in interpretation were reconciled through discussion.

2.5. Analysis: factors influencing park management policies

Text mining and network analysis were conducted using KH Coder (version 3; Boğusoğlu, 2024; Higuchi, 2016). In contrast with research focusing on user-generated content (Ishizuka, 2018; Ito and Sowa, 2010; Yasuhara and Liu, 2021; Yoshida, 2019), this study applies these methods to institutional narratives to describe urban park management discourses.

Data Preparation and Morphological Analysis

Transcripts were preprocessed by removing interviewer prompts and segmented into paragraphs as analytical units ("statements"). These were imported into the KH Coder for morphological analysis using the built-in ChaSen parser. Compound expressions (e.g., "Mizube-no-Sato," "Maruyama Park") were registered to preserve semantic clarity.

Frequency Analysis and Keyword Coding

For each park, the top ten terms by frequency were automatically identified. The three most salient keywords were examined within their sentence contexts and coded as Environmental Focus, User Focus, or Balanced orientations, based on keyword usage and paragraph meaning. Coding was performed manually, followed by iterative reflection and internal validation to ensure consistency across all statements.

2.6. Analytical Strategy for the 30-Level Framework

This study analysed the balance between environmental conservation and recreational use, by developing the analytical strategy illustrated in Fig. 1. This strategy is based on the premise that a park's tendency to adopt particular measures or policies across the 30 orientation levels is governed by the following three factors.

- External environmental factors
- Internal park conditions
- Management factors

Management orientations were categorised based on the combination of keyword frequency, contextual interpretation, and interview narratives, as supported by the content analysis described in 2.6.3.

2.6.1. Text mining methodology

KH Coder was used to conduct three types of analysis:

1. Frequency analysis to extract the most salient terms;
2. Keyword connectivity analysis using co-occurrence networks;
3. Categorisation of thematic orientation for each park.

Paragraphs were coded as "statements" and served as the basic unit for analysis. These statements were classified into one of the three management orientations, based on the usage and semantic context of the extracted keywords.

2.6.2. Construction of co-occurrence networks and connectivity analysis

To examine how keywords are semantically linked in interviews, co-occurrence networks were constructed for each park using the Jaccard coefficient:

$$\text{Jaccard}(A,B) = n(A \cap B)/n(A \cup B) \quad (2)$$

Here, A and B denote sets of paragraphs containing each term. The numerator indicates the number of co-occurrences, while the denominator represents all occurrences of either term.

The Jaccard coefficient ranges from 0 to 1 and is interpreted relative to the comparison.

Only pairs with coefficients ≥ 0.1 were retained to reduce visual clutter and focus on stronger links, as threshold values are not uniform and are selected to ensure visual readability (Nishio et al., 2016). Each network included nodes scaled by keyword frequency, edges weighted

by co-occurrence strength (with line thickness proportional to the Jaccard coefficient), and clusters identified by modularity to reveal thematic groupings. This enabled comparisons across parks, ensuring that distinctive management emphases were not lost in aggregation.

2.6.3. Content analysis

Content analysis was used to validate classifications derived from text mining. This method, widely applied in behavioural and management studies (Krippendorff, 2004; Zhang and Wildemuth, 2009), interpreted how managers described their practices in relation to ecological protection, user participation, and infrastructure development.

Statements were categorised into the three orientations defined in the Conservation and Utilisation Management Framework. This step enabled triangulation with keyword-based analysis and added interpretive depth to the understanding of park-specific orientations.

These three complementary approaches—frequency analysis, co-occurrence network analysis, and content analysis—were used in parallel to classify management orientations. Their relationships and analytical roles are summarised in Table 4.

3. Results and discussion

3.1. Results

3.1.1. Development of the conservation and utilisation management framework

Based on the concept of balancing environmental conservation and park use within the Conservation and Utilisation Management Framework, interview transcripts from all five parks were analysed to develop standardized descriptions for the 30-level management orientations. Table 5 summarises the results, in which the 30 ordered levels were further aggregated into three broader management orientations: Environmental Focus, Balanced, and User Focus.

For instance, Level 21 statements typically are interpreted as ecological commitments where eco-friendly natural materials are preferred, yet some artificial elements are permitted. Level 22 statements are interpreted as the exclusive use of natural materials to minimise infrastructure and maximise biodiversity conservation (Table 5).

3.1.2. Aggregated results of management orientations (Fig. 2)

The broader distribution of management priorities is visualised in Fig. 2, which shows the proportions of Environmental Focus, Balanced and User Focus by park, obtained by counting the classified descriptions for each management orientation across the five parks.

3.1.3. Aggregated results of operational dimensions (Fig. 3)

The proportion of operational dimensions by park is illustrated in Fig. 3. These dimension distributions were obtained by counting the classified descriptions for each operational dimension across the five parks.

Table 4
Analytical basis for interpreting top keywords by frequency and connectivity.

	Frequency Analysis	Connectivity (Co-occurrence) Analysis
Objective	Identify distinctions in management discourse and highlight themes characteristic or shared among parks.	Interpret contextual meaning of frequently co-occurring word pairs to reveal thematic associations and reasoning.
Unit of Analysis	Sentence containing the keyword.	Paragraph containing both co-occurring terms.
Interpretive Focus	Classify underlying management perspective as Environmental Focus, User Focus, or Balanced.	Explain relational significance of term connections and how they reflect specific management orientations.

Table 5
Results of the Conservation and Utilisation Management Framework.

	Operational dimension	User Communication	Categorisation of Statements Regarding Conflict Responses
UF	Experiential Activities		1.All activities permitted.
			2.Events held regardless of biodiversity impact (e.g., dining, music, sports).
	Hardware /Facility Development		3.Harmful substances or artificial facilities installed, minimal ecological care.
	Rules/Signage	Laissez-faire	4.Conservation rules exist but are not shown to users.
		Information Dissemination	5.Informational displays provided on local nature, without direct conservation role.
	Management Practices	Collaboration	6.Users involved in non-conservation management decisions (design, events, facilities).
			7.Harmful or dangerous species eradicated immediately.
			8.Woodland/waterfront management ignores biodiversity (pruning, dredging, etc.).
	Zoning/Land use		9.No buffer between natural areas and playgrounds/dining/facilities.
		Laissez-faire	10.No zoning for biological conservation.
BF	Experiential Activities		11.Activities offered for biodiversity learning and experience.
	Hardware /Facility Development		12.Facilities allow biodiversity observation while considering conservation.
	Rules/Signage	Regulations	13.Usage regulations include provisions for nature and biodiversity protection.
		Information Dissemination	14.Signage informs users of biodiversity-friendly use.
			15.Conservation rules exist and are communicated, but no strict access limits.
	Management Practices	Collaboration	16.Collaborative management with users/residents/organisations.
			17.Monitoring conducted; rare species information shared with users when appropriate.
	Zoning/Land use		18.Buffer zones created between conservation and user spaces.
		Regulations	19.Conservation areas normally closed, opened for events/ activities.
			20.Restricted access to paths in designated conservation zones.
EF	Hardware /Facility Development		21.Eco-friendly natural materials used where possible.
			22.No harmful substances or artificial facilities; structures built from natural materials (limited administrative exceptions permitted).
	Rules/Signage	Information Dissemination	23.Conservation policy exists but not visibly communicated or enforced.
			24.Users asked, out of courtesy, to avoid harmful behaviour.
			25.Conservation rules exist; warnings via signage, not direct confrontation.
			26.Patrols and direct warnings enforce conservation rules.
	Management Practices		27.Management led by experts or follows expert advice.
			28.Monitoring conducted; rare species info withheld for protection.

(continued on next page)

Table 5 (*continued*)

	Operational dimension	User Communication	Categorisation of Statements Regarding Conflict Responses
	Zoning/Land use	Regulations	29.Temporary access restrictions for biological protection or restoration. 30.Areas permanently closed to non-management/non-research entry.

*UF: User Focus, BF: Balanced Focus, EF: Environmental Focus

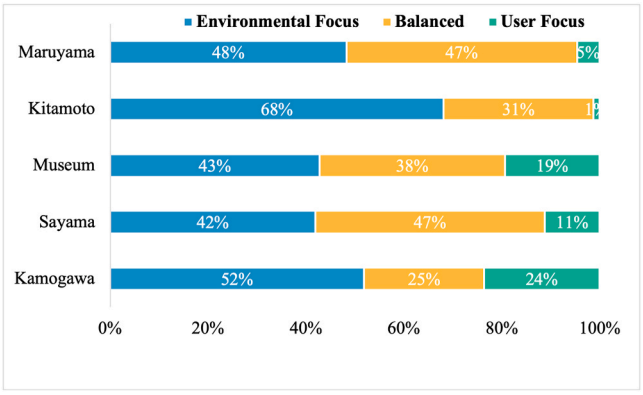


Fig. 2. Proportion of Environmental Focus, Balanced and User Focus by park.

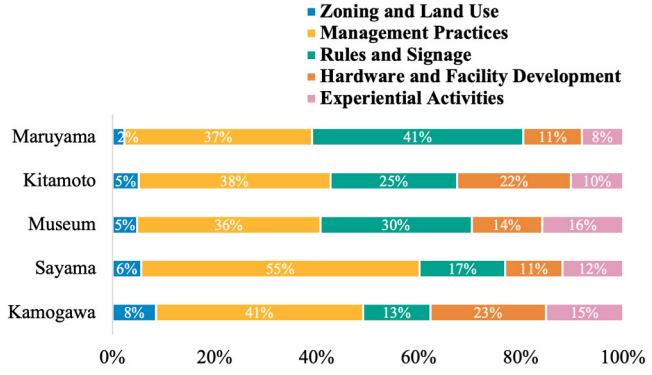


Fig. 3. Proportion of operational dimensions by park.

3.1.4. *Results of keyword frequency analysis (Table 6)*

Keyword frequency analysis was conducted using KH Coder. The top three keywords interpreted under the theme of urban park planning for each park, along with their frequencyies, were obtained from the co-occurrence networks and presented in Table 6.

3.1.5. *Results of co-occurrence network and Jaccard coefficients (Fig. 4)*

Co-occurrence network analysis was conducted using KH Coder to generate diagrams, with Sayama featured as a representative case (Fig. 4), as well as Figs. A.1–4 (partially in the Appendix). From these networks, the top-ranked Jaccard coefficient pairs representing the connectivity for each park were obtained, as presented in the Keyword Connectivity Interpretation Table (Table A.3a-e).

3.2. *Discussion*

3.2.1. *Comparison of orientation profiles across parks*

To evaluate and compare the proportions of management orientations across contexts, the framework's classification results were aggregated for each site. As shown in Fig. 2, the distributions of Environmental Focus, Balanced and User Focus vary among the parks.

Table 6
Top three keywords interpreted under the theme of urban park planning.

Park Name	Top Three Words (Frequency)	Urban Park Planning	
		Discussion of Urban Park Policy	Discussion of Maintenance Policy
Kamogawa	People (40) Park (35) Meadow (18)	Meadow-based play; respect for local nature and heritage; participatory zoning and rule-making	Meadow creation & upkeep; community workshops; feature projects (e.g., dragonfly pond)
Sayama	Centre (52) Management (52) People (50)	Satoyama conservation; resident–volunteer councils; expert-advisor guidance	Cooperative maintenance; consensus-building; resident feedback integration
Museum	People (37) Management (35) To Create (30)	Citizen involvement in conservation; volunteer partnership; collaborative restoration	Biodiversity & visitor safety; area-specific monitoring; event-based education
Kitamoto	Park (51) Nature (38) Animals (29)	Nature-oriented design; history-informed; biodiversity protection; integrate adjacent biotopes (Arakawa)	Wetland & woodland conservation; scientific monitoring; restricted access
Maruyama	Park (104) People (37) Nature (35)	"Harmony of Water & Greenery"; water-cycle management; protect water-edge landscapes	Kaibori pond restoration; adaptive water-use rules; controlled water-based activities

Note: Full details of Table 6 are provided in Appendix Table A.1.

Kamogawa (52% Environmental Focus, 24% User Focus) is characterised by clear spatial zoning that separates conservation and recreational areas. Interview evidence indicates community-based management arrangements, and the park is adjacent to residential areas with strong recreational demand.

In contrast, Kitamoto, with a dominant Environmental Focus (68%) and minimal User Focus (1%), adopts an exclusionary management style. The park lies within a sensitive wetland corridor and was designed with biodiversity conservation as a primary aim. The interviews indicate relatively limited direct demand for recreation.

Both Sayama and Maruyama display higher Balanced orientations (47%), but their underlying strategies differ. Sayama emphasises sustainable public engagement, often partnering with schools and volunteers for restoration projects, yet deliberately limits the development of user-focused facilities to safeguard ecological priorities. Maruyama, which was formerly more user-oriented, has shifted towards conservation by phasing out activities incompatible with ecological goals. The interview accounts refer to restoration progress (such as the revitalisation of aquatic habitats) and changing policy priorities alongside this shift.

Museum, with its extensive natural areas and low visitor density, combines user amenities (such as interpretive signage and observation events) with a minimum-intervention approach, leveraging flexible spatial zoning to protect biodiversity and accommodate moderate public use.

These results vary across ecological context, park history, user demands, and institutional priorities. They are discussed in relation to integrating quantitative findings with qualitative insights for understanding urban park governance. The Keyword Connectivity Interpretation Table is a reference for identifying park management priorities.

3.2.2. *External environmental factors*

External factors, such as surrounding ecosystems, proximity to residential areas, and agricultural land, are important for understanding management decisions.

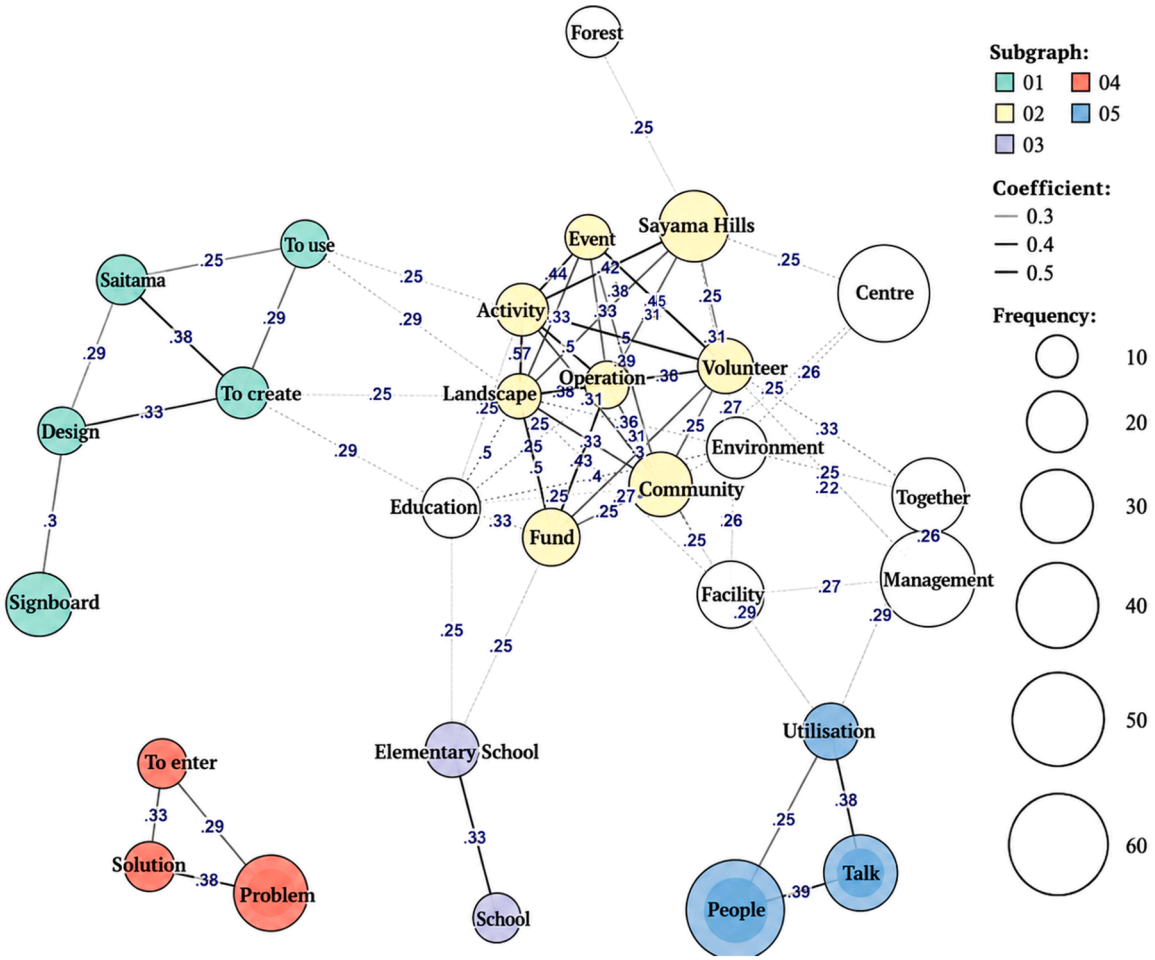


Fig. 4. Co-occurrence network diagram for Sayama.

- Surrounding Ecosystems: Kitamoto, designed from its inception as an urban park with a primary focus on nature conservation, is located along the Arakawa River Basin ecological network and adjacent to the Arakawa Biotope, a pioneering large-scale restoration project. Fig. 2 shows a strong conservation-oriented profile (68% Environmental Focus; 1% User Focus) The site is situated within approximately 50 ha of wetland habitat under conservation management.
- Proximity to Residential Areas: Kamogawa and Sayama are both situated near residential communities. Kamogawa has a significantly notably higher proportion of User Focus (24%) (Fig. 2). Sayama, meanwhile, shows the highest proportion of Balanced orientation among all parks (47%), alongside a substantial share of Environmental Focus (42%) and a moderate User Focus (11%). Interview evidence indicates diverse expectations from local communities and reported efforts to harmonise public use and ecological care.
- Agricultural Proximity: Kitamoto is adjacent to agricultural land, and interview accounts indicate the inflow of nutrient-rich runoff that poses ongoing challenges for wetland conservation and water quality. Wetland protection and management are repeatedly described as central priorities. As presented in the top word pairs in the Keyword Connectivity Interpretation Table (Table A.3d), such as "Maintenance × Wetland" and "Organisms × Problem", management efforts focus on sustaining wetland functions and addressing related ecological issues. The frequent pairing of "Caution × Signboard" involves the park's use of rules and signage to regulate visitor behaviour and minimise further impacts on sensitive habitats.

These spatial and social environmental influences are discussed in relation to the need for adaptive, place-based management.

3.2.3. *Internal environmental factors*

Internal factors, including ecological naturalness, spatial scale, and land ownership, are discussed in relation to operational dimensions (Fig. 3).

- Naturalness and Scale: Museum, located deep in the Sayama Hills, has the highest ecological naturalness. Its large area permits participatory programmes and signage, while concentrating visitor facilities (16%) in less-sensitive zones. Staffing is limited; management relies on rules and signage (30%). The park is remote and experiences relatively low visitor pressure, a configuration that is discussed alongside a balance between conservation and use.

  By contrast, Kitamoto, a compact valley park, is prone to disturbances and prioritises conservation, with high shares of management practices (38%) and rules (25%).

  Urban parks such as Maruyama and Kitamoto depend heavily on signage (41% and 25%) under strict environmental policies. In Maruyama, signage is used to reinforce gains in aquatic restoration in a dense urban setting.

  Kamogawa, the smallest and most linear site, applies zoning (8%) and facility investment (23%) to separate conservation from recreation, an adaptation to its residential adjacency.

  Sayama, a large regional park, devotes the highest share to management practices (55%). Zoning distinguishes accessible forests from unmanaged woodlands, balancing community use and habitat

protection. As emphasised by its manager, zoning is necessary for managing the diverse landscape.

- Ownership: Sayama has fragmented tenure, with private plots interspersed with Foundation-owned parcels. The Foundation's "National Trust" has conserved 52 lots, a process that requires constant negotiation among landowners, governments, residents, and the Foundation. Interviews describe a polycentric arrangement with ongoing negotiation. Fig. 3 shows a high share of management practices (55%).

Kitamoto's unified public ownership is reported alongside centralised, conservation-led management. Maruyama and Kamogawa, also public, benefit from streamlined administration but risk limited stakeholder diversity. Museum, with extensive public land, relies on visitor self-regulation through zoning and signage.

Overall, fragmented tenure and collaboration are frequently reported together, while unified tenure is frequently reported with administrative efficiency. These patterns are highly context-dependent.

●Experiential Activities: Lower proportions of Experiential Activities at Kamogawa (15%) and Maruyama (8%) are discussed in relation to classification overlap within the analytical framework. Many experiential interactions in these parks are categorised as Environmental Education or Management Practices rather than as standalone experiential programmes; thus, the data do not necessarily indicate fewer activities. This is interpreted in light of the importance of integrating quantitative results with qualitative insights to accurately capture the complexity of on-site management.

3.2.4. *Management factors*

Management factors, including stakeholder involvement, volunteer participation, collaboration, policy orientation, and professional expertise, are evident in management strategies and outcomes across the five parks. This section synthesises evidence (Table 6; Tables A.4-A.6) and interview data to show how each management factor is implemented at each site and how internal and external conditions relate to cross-site variation.

- Management Perspectives on Stakeholder Involvement and Expertise: Kitamoto exemplifies a conservation-first approach, prioritising biodiversity protection through expert-driven strategies and professional ecological knowledge. Interviews emphasise targeted conservation goals and relatively compact areas, as well as the need for specialist input to achieve their objectives (Table 6; Tables A.5 and A.6).

  Museum, though also ecologically oriented, adopts a minimal intervention approach (Table A.5), relying heavily on skilled volunteers to maintain ecological quality (Table 6; Table A.4) and deliver curated educational activities, such as the annual "Green Forest Festa" (Table A.6), across a much larger park area.

  Maruyama adopts a philosophy of "restoration through community participation." In this model, trained local volunteers (such as mizube-mamori) serve as intermediaries between ecological specialists and general park users, playing a key role particularly during restoration initiatives, such as pond regeneration (Kaibori) (Table 6; Table A.5). These projects routinely attract significant involvement from both trained and general volunteers, who facilitate communication, guidance, and on-the-ground collaboration. Such collaborative efforts are reinforced by management's emphasis on phased environmental education and broad-based citizen engagement (Tables A.5 and A.6), to sustain restoration and conservation objectives over time.

  These variations are interpreted in light of how internal priorities, such as ecological value, park history, or ownership complexity, and surrounding social conditions collectively shape management philosophies. Conservation-oriented parks tend toward expert or volunteer-driven approaches, while socially complex or urban parks favour participatory and adaptive perspectives. This is discussed in relation to the need for context-sensitive governance models in urban park management, ensuring that management approaches respond to both ecological conditions and social realities.
- Policies: The implementation of these philosophies yields distinctive policy frameworks tailored to each park's context. Kitamoto implements strict ecological rules, habitat-sensitive infrastructure, and restricted access, with daily maintenance and intervention guided by expert consultation and scientific monitoring (Table 6; Tables A.5 and A.6). The museum maintains clear, area-specific rules and minimal intervention, guided by a small but skilled volunteer group and selectively offered educational programming (Table 6; Tables A.4-A.6).

  Maruyama employs explicit signage and adaptive policies to manage aquatic habitats, strictly prohibiting fishing and other water-based recreational activities to safeguard restoration outcomes achieved through pond restoration (Kaibori). The stringent regulations are interpreted as intended to limit potential ecological disturbances from fishing, thereby clearly communicating conservation-focused management to visitors, and to support the maintenance of the ecological gains from periodic Kaibori activities (Table 6; Tables A.4 and A.6). Kamogawa's policies are shaped by participatory rule-making, consensus-based decision-making, and collective safety management, with policy adjustments responding flexibly to user input gathered through workshops and open dialogue (Table 6; Tables A.5 and A.6). Sayama employs ongoing consultation, collaborative maintenance, and proactive responsiveness to resident and user needs, with adaptive management grounded in regular dialogue among managers, residents, and expert advisors (Tables A.4 and A.5).

  Parks adjacent to residential areas (Sayama, Kamogawa) typically adopt more participatory and adaptive policies, while parks with higher ecological sensitivity or remoteness (Kitamoto, Museum) enforce stricter, conservation-oriented rules. Reflecting its urban location and mission-driven history, Maruyama uniquely balances restoration and recreation through policy models tailored to both ecological and social realities.
- Collaborators and Expertise: Collaboration structures and the mobilisation of expertise further distinguish the five parks. Sayama demonstrates the most embedded, multi-level collaborative network, with managers, residents, volunteers, and external experts participating through community councils, joint maintenance, and regular consultation (Table 6; Tables A.4-A.6). Interviews describe this structure in relation to fragmented landownership and complex social settings.

  The museum is maintained by a small, highly committed volunteer core, supplemented by specialist partners (Table 6; Tables A.4 and A.5), and volunteers are formally acknowledged for their essential roles. Maruyama features broad citizen participation, with volunteers, local groups, and city agencies jointly coordinating efforts on river restoration, invasive species control, and environmental education (Tables A.4 and A.6). Kitamoto, by contrast, relies on professional ecological advice, scientific monitoring, and formal management contracts, with less dependence on general volunteer networks. Kamogawa integrates structured management and regular informal user involvement, blending participatory mechanisms with management-led safety and ecological care (Table 6; Tables A.5 and A.6).

  These patterns are interpreted in the context of broad, participatory collaboration, which is more common where ownership is fragmented or urban access is high (Sayama, Maruyama, Kamogawa). At the same time, expert networks and focused volunteer teams dominate in ecologically sensitive or less accessible parks (Kitamoto, Museum). Thus, collaboration reflects both the physical and institutional context, ultimately informing each park's governance capacity and adaptability.

- Theoretical and Methodological Reflection: This cross-park analysis is conducted in light of how management orientations and policy frameworks are shaped by site-specific ecological and social conditions and dynamically co-constructed through adaptive, multi-actor collaboration. By integrating keyword data (Table 6; Tables A.4-A.6) and interview evidence, this study provides an adaptable framework for evaluating management strategies in urban green spaces. The diversity of approaches observed, from expert-driven ecological sanctuaries to participatory, community-based models, is discussed in relation to the necessity of context-sensitive governance. This 30-level framework provides a unified, standardised expression of diverse management approaches, enhancing cross-case referential value. Its practical relevance is supported by interviews with experienced professionals possessing extensive expertise in biodiversity conservation and habitat management, whose knowledge has been applied to urban park contexts. The interviews were conducted in 2019–2020. Crucially, Japan's National Biodiversity Strategy 2023–2030, adopted following COP15, has positioned urban parks and green spaces as one of the relevant settings for advancing Nature Positive objectives. While on-site conservation initiatives remain limited across municipalities, drawing on the experience of these professionals externalises scarce, practice-based conservation management knowledge, providing adaptable reference points for local authorities responding to emerging biodiversity conservation goals.

Theoretically, this study demonstrates that, in urban parks aiming to actively restore nature, as well as in peri-urban parks seeking integrated conservation with surrounding ecosystems, conflicts may arise between the free use of parks by urban residents and the protection of biodiversity. In such situations, the findings suggest that an appropriate level of conservation emphasis can be adjusted and adjusted in response to internal and external environmental conditions and management constraints. In practice, the research underscores the critical importance of adaptive management strategies that integrate both biological adaptive management—the responsive and flexible adjustment of ecological processes—and social adaptive management, which entails ongoing negotiation and mediation among diverse stakeholder values and sometimes conflicting demands.

As substantiated by Table 6 and A.4–6 and interview testimony, the diversity of management perspectives, policy frameworks, and collaborative structures helps explain the observed variation in park management strategies. Kitamoto and Museum exemplify expert-driven, conservation-focused management, especially where ecological value or remoteness shapes priorities. Maruyama, Kamogawa, and Sayama demonstrate the importance of participatory, adaptive governance in parks situated within urban or socially complex contexts. Ultimately, the alignment between management vision, policy design, and collaborative practice is key to assessing each park's capacity to balance environmental conservation with public use, offering valuable lessons for urban park governance globally.

More broadly, the analysis demonstrates that external environmental factors, internal park conditions, and management factors do not operate in isolation. Rather, they interact within each park, collectively informing management orientations and specific strategies. This multilayered, context-dependent interplay is discussed in relation to the need for integrative and adaptive approaches in urban park governance, ensuring that management practices remain responsive to the complex realities of each site.

3.2.5. Co-occurrence network and connectivity analysis

This section explains how park management reconciles biodiversity conservation with public use by applying co-occurrence network analysis. In each park, the top-ranked Jaccard coefficient pairs not only are presented as the most salient semantic associations in management discourse but also encapsulate the management focus and unique character of that park (Fig. 4; Figs. A.1–4). The Keyword Connectivity Interpretation Table with Jaccard coefficients provides information for interpreting relationships among objects, ideas, purposes, and challenges, thereby informing the understanding of management orientations.

Kamogawa (Fig. A.1), a relatively small, linear park in proximity to residential areas, emphasises participatory management through community-led workshops, as indicated by the strong association 'organisation × workshop' (Jaccard coefficient: 0.33, ranked second). As observed in the Keyword Connectivity Interpretation Table (Table A.3a), these workshops are described as direct platforms for dialogue between residents and management and for collective decision-making. Interview evidence indicates robust stakeholder involvement and consensus-based discussion of resident needs and ecological priorities.

Sayama (Fig. 4) prominently connects the associations "landscape × activity" (Jaccard coefficient: 0.57, ranked first), "landscape × education" (0.50, ranked second), and "event × volunteer" (0.45, ranked third), as observed in the Keyword Connectivity Interpretation Table (Table A.3b). The network places "landscape" at the centre of these links. Interviews describe an emphasis on stewardship of Satoyama landscapes, with schools and volunteers participating in events and educational activities. The prominence of "landscape" is interpreted in light of targeted community-based educational activities and the preservation of traditional rural ecological settings.

Museum (Fig. A.2) features the keyword pair "to appear × utilisation" (Jaccard coefficient: 0.35, ranked first), Interview evidence indicates spontaneous vegetation growth, such as seasonal plant emergence, as opportunities for visitors to observe and learn about natural dynamics. While the network diagram highlights the statistical strength of this connection, the connectivity interpretation table (Table A.3c) provides its management context, indicating that this co-occurrence reflects repeated discussions linking seasonal natural changes with visitor experiences.

Regarding the keyword pair "nature × to create" (Jaccard coefficient: 0.33, ranked second), interview data indicates a commitment to maintaining the site's existing natural conditions and a reluctance to introduce new roads or facilities. A conflict between facility construction and the preservation of the park's ecological integrity is noted, with priority given to protecting the site's natural state over enhancing visitor convenience through additional infrastructure. This management philosophy is fundamental to the museum's identity as an ecological sanctuary and sets it apart from more interventionist or user-oriented parks in the study.

Kitamoto (Fig. A.3) is interpreted as a distinct, restoration-oriented management philosophy. The keyword pairs "maintenance × wetland" (Jaccard coefficient: 0.29, ranked first) and "organisms × problem" (0.27, ranked third) are particularly prominent in the Keyword Connectivity Interpretation Table (Table A.3d). The interview content indicates that wetland ecosystem management, including seasonal maintenance to sustain biodiversity and ecological stability, is a core focus, alongside active responses to biodiversity issues, such as invasive alien species and water quality.

Unlike Museum and Sayama, which favour preserving natural conditions through minimal intervention, Kitamoto actively identifies and responds to biological and ecological challenges, such as the management of invasive species and water quality, which are closely monitored and managed by specialists. This is interpreted in light of Kitamoto's restoration-oriented management philosophy, in contrast with the non-interventionist approach of Museum and Sayama.

Maruyama (Fig. A.4) prominently features the keyword pairs "nature × park" (Jaccard coefficient: 0.43, ranked first), "convenience × to consider" (0.42, ranked second), and "location × utilisation" (0.42, ranked third), all emphasised in the Keyword Connectivity Interpretation Table (Table A.3e). Interview evidence indicates an ecological restoration and observation-focused philosophy, including pond

draining (Kaibori), and frames convenience and safety in relation to facilitating safe nature observation (e.g., lighting and open pathways to prevent accidents). Interviews also reported reservations about leisure-oriented use and a preference to restrict facilities and programming to ecological observation.

In sum, these networks, anchored by quantitative co-occurrence rankings, a Keyword Connectivity Interpretation Table, and interview evidence, are presented as word pairs that associate with each park's management emphases and local contexts (Fig. 4; Figs. A.1–4). The diversity of semantic associations further reflects how geographic location, park history, stakeholder structure, and local needs shape tailored strategies. These patterns are examined in relation to the value of the connectivity interpretation table as a tool for cross-case comparison and context-sensitive, adaptive management in urban park governance. More broadly, these findings underscore the value of mixed-method, cross-case analysis for advancing both the theory and practice of adaptive green space governance.

At a strategic level, this keyword connectivity provides insights into managers' repeated emphasis and thematic clusters,as reflected in the relationships among frequently connected terms. These patterns help interpret how different management orientations—represented in this study as Environmental Focus, Balanced, and User Focus—are prioritised and combined within specific park contexts. The results suggest that managers do not rely exclusively on a single orientation; instead, they flexibly combine multiple orientations (Fig. 2) depending on ecological conditions, user relationships, and challenges, reflecting characteristics consistent with adaptive management approaches.

Moving forward, future research involving a larger number of parks could use these recurrent keyword networks to examine further the relationships between park environmental conditions and prioritised management measures, the interactions between surrounding community contexts and management constraints, and the processes through which adaptive conservation practices engage urban communities and generate broader social benefits.

Such adaptive processes may provide opportunities to experimentally involve urban residents in biodiversity-oriented management practices.

3.2.6. Land-use context and stakeholder involvement

External factors, such as ecological settings and residential contexts, generally steer parks toward either conservation-oriented or community-oriented directions. Internal factors related to land ownership patterns play a particularly decisive role in shaping forms of environmental governance. The extensive area and fragmented private ownership observed in Sayama require collaborative governance. In contrast, the integrated public ownership of a relatively compact, geomorphologically fragile wetland landscape, as seen in Kitamoto, enables an expert-led, conservation-oriented, centralised management model.

Management perspectives and stakeholder involvement, mediated through land ownership structures, generate diverse governance arrangements ranging from highly community-driven collaboration, as in Sayama and Kamogawa, to volunteer-supported models, as exemplified by Museum and Maruyama. In Kamogawa, the influence of the adjacent residential areas were associated with expert-led collaboration largely limited to neighboring communities.

3.2.7. General discussion

Taken together, these findings indicate that environmental management in sustainable parks depends not only on technical interventions but also on adaptive and collaborative communication with users. A main contribution of this study is the development of a conceptual tool to guide environmental governance in parks across diverse local contexts.

This study identifies the various barriers to, and countermeasures for, nature conservation efforts implemented through the institutional framework of public parks. The analysis of five cases in Saitama Prefecture, Japan, inevitably, has limited generalizability.

For example, two of the parks are regional, with extensive areas largely comprising privately owned land with highly fragmented ownership. This condition presents challenges to advancing environmental conservation activities and promoting recreational use, reflecting land tenure patterns historically rooted in Japan.

At the same time, however, the challenge of balancing conservation and use when artificially developed urban spaces or agricultural lands with high ecological restoration potential are influenced by their proximity and accessibility to urban residents is not unique to Japan. If further examined from the perspective of parks as green infrastructure that delivers ecosystem services, this issue could facilitate the sharing of common challenges at the international level.

Additionally, longitudinal perspectives on "Balanced" cases and on the long-term sustainability of such balance, as evaluated by managers, remain unresolved. More critically, further research is needed to examine the lessons these five parks, with their advanced initiatives, have drawn from the continuation and evolution of their practices and from adaptive management responses. Furthermore, incorporating user perspectives and scientifically examining the biophysical and social conditions within and around parks will be essential for optimizing location-sensitive conservation strategies within urban planning and park management.

4. Conclusions

Drawing on interview data, this study developed a 30-level Conservation and Utilisation Management Framework and used it to examine how five parks in Saitama Prefecture balance environmental conservation and recreational use. Additionally, interview transcripts were analysed using an integrated approach combining keyword frequency analysis with co-occurrence network analysis, based on the premise that speech characteristics reflect each park's prioritised environmental management orientations and challenges.

The extracted keyword sets were categorised into Environmental Focus, Balanced Focus, and User Focus using the 30-level framework, enabling a proportional analysis of each park's orientation. These findings were cross-validated against independently documented park conditions, including internal park conditions and external environmental contexts. The results revealed several points of convergence between speech patterns and internal/external environmental and management factors, suggesting that the tendencies somewhat align with each park's orientation toward Environmental Focus, Balanced Focus, or User Focus.

CRediT authorship contribution statement

Junying Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kiyotaka Fukahori:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to improve the readability, clarity, and grammatical accuracy of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Sayama Hill Flora & Fauna Interaction Centre / Saitama Green Forest Museum / Saitama Nature Study Centre at Kitamoto Nature Observation Park / NPO eco-works at Kamogawa Mizubenosato Park / NPO eco-works at Ageo Maruyama Park

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ufug.2026.129577.

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Glossary

Satoyama: A traditional Japanese rural landscape characterised by a continuous mosaic of secondary forests, grasslands, and wetlands.

Kaibori: A traditional Japanese pond-draining practice in which water is removed from ponds or marshes, mud is cleared, aquatic organisms such as fish are harvested, and the pond bed is left to dry in the sun.